YASHIKA SINGH

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Summary

AI/ML Engineer with around 4 years of experience building end-to-end AI products, from LLM-powered applications to ML models. Skilled in RAG, prompt engineering, and multimodal AI, with production experience delivering projects at Product Manager Accelerator under Dr. Nancy Li, as Lead AI Engineer at Columbia University's Summer Startup Program and as AI/ML Technical Lead for a stealth fashion startup in New York, owning the end-to-end AI architecture and roadmap. Selected for the prestigious MOVE Program by Handshake, recognizing top Master's students for leadership and innovation in technology.

Education

Yeshiva University, Katz School of Science and Health, New York, NY
Master of Science in Artificial Intelligence, GPA: 3.8/4.0 — May 2025

Galgotias College of Engineering and Technology, Greater Noida, India
Bachelor of Technology in Computer Science, CGPA: 3.5/4.0 — Aug 2021

Work Experience

AI/ML Technical Lead, Stealth Startup - Fashion, New York — Dec 2025 - Present

- Leading all AI/ML initiatives, owning end-to-end technical architecture, model selection, and implementation strategy for personalization, recommendations, and automation.

Model Validation Expert Fellow, Handshake, San Francisco, CA (Remote) — Sep 2025 - Dec 2025

- Collaborated with leading researchers to refine the capabilities of Large Language Models (LLMs), contributing to testing, evaluation, and contextual improvements in AI systems.
- Inducted as a Star Fellow, a distinction reserved for top-performing fellows recognized for creating the highest-quality prompts and contributions to AI research and evaluation.

Lead AI Engineer, Columbia University Summer Startup Program, New York — July 2025 - Aug 2025

- Collaborated with Columbia University student founders on a fintech startup initiative, providing AI leadership from concept to MVP and achieved a successful launch within a short timeframe, including the end-to-end design of RAG pipelines, LLM integrations, and product-facing dashboards.
- Lead the end-to-end development of Retrieval-Augmented Generation (RAG) pipelines with integrated LLMs, handling both backend architecture and frontend user experience using Django framework.
- Delivered client-facing dashboards and technical documentation, and drove technical decision-making and roadmap planning for the MVP launch.

AI/ML Engineer Intern, Product Manager Accelerator, Boston, MA (Remote) — Jan 2025 - Nov 2025

- Designed and deployed AI-driven interview simulation workflows for Product Manager Accelerator's proprietary platform, directly under the leadership of founder Dr. Nancy Li, leveraging RAGs and LLMs (OpenAI, Claude) to generate context-aware product management and behavioral assessments.
- Applied NLP techniques to develop dynamic prompt engineering, follow-up logic, and structured candidate evaluation.
- Improved real-time response performance by 70% by optimizing model pipeline and collaborated with cross-functional teams to migrate business logic into production.

Data Scientist, Axtria, Noida, India — Aug 2021 - Dec 2023

- Optimized Python scripts to streamline data processing pipelines, improving performance and reducing manual effort.
- Developed and maintained ETL pipelines using Matillion, SQL and Python.
- Automated complex data processing workflows using DAGs in Apache Airflow, reducing manual intervention by 40% and enabled autonomous operations in dynamic environments.
- Integrated MLOps best practices with CI/CD pipelines to automate model training, deployment and monitoring on Databricks, ensuring scalability and reliability.
- Optimized automated pipelines in Azure Data Factory and Apache Airflow, reducing data processing errors and improving overall efficiency.
- Created interactive dashboards and data visualizations using Power BI and Tableau to deliver insights and support decision-making for internal stakeholders.

Projects

Drowsiness Detection for Drivers: [GitHub Link]
Built a CNN-based, real-time system using TensorFlow to detect driver fatigue and improve road safety with validation accuracy around 95%.

Detection of Cardiomegaly in Dogs: [GitHub Link]
Created a deep learning model using custom CNN for X-ray analysis and automating veterinary diagnostics with an accuracy of 76.75%, surpassing the benchmark set by VGG16 (75%).

Skills and Certifications

Programming Languages: Python, C, C++, JavaScript, SQL
Developer Tools/Cloud: Apache Airflow (DAGs), Azure Data Factory, AWS, Google Cloud Platform (GCP), Django, VS Code, Git, TensorFlow, Streamlit, Tableau, Power BI, PyTorch, Computer Vision, Anaconda, Jupyter, Google Colab, Pandas, Scikit-learn, Matplotlib, Salesforce Data Cloud, Databricks
Skills: Natural Language Processing (NLP), Generative AI, Large Language Models (LLM), Prompt Engineering, Retrieval-Augmented Generation (RAG), Convolutional Neural Networks (CNN), Data Analysis, Machine Learning, Deep Learning, Automatic Speech Recognition (ASR), Predictive Modeling, Exploratory Data Analysis (EDA), Whisper, MySQL, PostgreSQL, Extract-Transform-Load (ETL), HTML5, CSS